

Sprint Plan – Sprint 2

Project: AI/ML Incident Response Optimization

Team: Latifa Osman, Eric Chen, Nishaaj Islam

1. Product Vision

The goal of this project is to develop a data-driven system that helps IT leaders monitor incident performance, identify inefficiencies, and improve decision-making. By using cleaned ServiceNow data, dashboards, and predictive analytics, the system will reduce incident resolution time, improve SLA compliance, and optimize resource allocation across support teams.

2. Product Backlog

1. Incident Volume Dashboard (Completion)

- **Priority:** High
- **Story Points:** 2
- **User Story:**
As a CIO, I want to see incident volume over time so that I can understand workload trends.
- **Acceptance Criteria:**
 - Shows daily, weekly, and monthly incident counts
 - Accurate based on cleaned dataset
 - Clear and readable visualization
- **Reasoning:** Work was already started in Sprint 1, so it requires low effort to complete.

2. SLA Adherence Dashboard (Completion)

- **Priority:** High
- **Story Points:** 3

- **User Story:**
As a manager, I want to monitor SLA compliance so that I can identify performance issues and violations.
- **Acceptance Criteria:**
 - SLA compliance percentage calculated
 - Violations identified
 - Visualization included
- **Reasoning:** Partially completed in Sprint 1, but requires additional logic and visualization.

3. Assignment Group Performance Analysis

- **Priority:** High
- **Story Points:** 4
- **User Story:**
As a CIO, I want to compare resolution time across assignment groups so that I can identify inefficient teams.
- **Acceptance Criteria:**
 - Average resolution time per group calculated
 - Comparison visualization created
- **Reasoning:** New feature that requires grouping and analysis, making it more complex.

4. Category-Based Analysis

- **Priority:** Medium
- **Story Points:** 3
- **User Story:**
As a manager, I want to analyze incidents by category so that I can identify common issue types and patterns.
- **Acceptance Criteria:**
 - Top categories identified
 - Resolution time by category calculated
 - Visualization included

- **Reasoning:** Moderate complexity using grouping and aggregation.

5. Incident Lifecycle Visualization

- **Priority:** Medium
- **Story Points:** 3
- **User Story:**
As a stakeholder, I want to visualize the lifecycle of incidents so that I can understand how incidents move through the system.
- **Acceptance Criteria:**
 - Shows stages (opened → resolved → closed)
 - Displays time spent between stages
- **Reasoning:** Requires transformation of timestamp data.

6. Data Validation Enhancements

- **Priority:** Low
- **Story Points:** 2
- **User Story:**
As a data analyst, I want to validate the dataset further so that analysis results remain accurate.
- **Acceptance Criteria:**
 - Additional checks for missing or inconsistent data
- **Reasoning:** Minor improvements with low complexity.

3. Velocity Measurement

In Sprint 1, the team completed the following:

- Data cleaning and validation (4 points)
- Resolution time analysis (3 points)
- Partial SLA tracking and dashboard work (2 points)

Total completed = 9 story points

Velocity = 9 story points per sprint

Velocity was calculated by summing the story points of completed user stories. Partial work was estimated proportionally based on completion level.

4. Product Roadmap

- **Sprint 1:** Data cleaning and core analytics (completed)
- **Sprint 2:** Expand analytics and complete dashboards
 - Finish dashboards
 - Add group and category analysis
- **Sprint 3:** Predictive modeling and recommendations
 - Build prediction model
 - Generate optimization strategies
 - Add export and filtering features

5. Sprint Goal

The goal of Sprint 2 is to complete the core dashboards and expand the analytics by adding group-level and category-based insights to better identify inefficiencies in incident management.

6. Sprint Backlog

User Story	Story Points
Incident Volume Dashboard (complete)	2
SLA Dashboard (complete)	3
Assignment Group Analysis	4
Category Analysis	3

Lifecycle Visualization	3
-------------------------	---

Total = 15 story points

This exceeds the team's velocity of 9, ensuring a productive sprint while still being achievable.

Justification:

These user stories were selected because they directly support the sprint goal of expanding analytics and improving insights. Completing unfinished work ensures continuity, while new features provide deeper analysis for decision-making.

7. Sprint Tasks

Incident Volume Dashboard

- Clean and format date fields
- Aggregate incidents by day, week, and month
- Create line chart visualization

SLA Dashboard

- Standardize SLA values
- Calculate compliance rate
- Identify violations
- Create visualization

Assignment Group Analysis

- Group data by assignment group
- Calculate average resolution time
- Build comparison chart

Category Analysis

- Group data by category
- Count incident frequency

- Calculate average resolution time
- Create visualization

Lifecycle Visualization

- Calculate time between incident stages
- Build timeline or flow visualization